

EP03 — How does the internet reach your phone?

ENGLISH

01 Title

How does the internet actually reach your phone? Here's the spoiler that surprises most people: it's almost **never** wireless. Let me show you the real journey your data takes.

[CUE] Hold the title ~2s. Curious, "let me reveal a secret" tone.

02 The Hook

You tap a video here in India, and it loads from a server sitting in America — in a few **milliseconds**. No wire ever touches your hand. So how does that data cross the whole planet, that fast? Let's follow it.

[CUE] Energy up. Gesture "across the world" with your hands.

03 Today's question

So today's question: when you open a website, what **physically** travels — and how **far** does it really go? Let's find out.

[CUE] Slow down. Let "how far" hang for a second.

04 Idea 1 — WiFi is the last hop

Idea one. The word "WiFi" fools everyone. WiFi is only the **last few metres**. Your phone talks to your router using radio waves — but that wireless hop is tiny, just to the little box on your wall. After the router, there is **no more air**.

[CUE] Point: phone → router. Emphasise "just to the box on your wall".

05 Idea 2 — It's cables

Idea two. From your router onward, it's **wires** — lots of them. Fibre-optic cables run across your city, across countries, and then giant cables lie on the **ocean floor**, connecting the continents. Around ninety-nine percent of the internet is physical cable. Not air. Not satellites.

[CUE] Trace the chain: router → city → undersea → data centre.

06 Idea 3 — Finding the server

Idea three. Your data isn't wandering randomly — it's going to one **exact computer**. You type a name like youtube-dot-com, and that name maps to an **address** — an IP, like a phone number for computers. Your request travels to that address, finds the right server in some building, and the reply comes **back** to your phone.

[CUE] Keep it light — we go deeper into packets in the next episode.

07 Idea 4 — Speed of light

Idea four — why does it feel **instant**? Because fibre carries your data as pulses of **light**. Mumbai to London is about seven thousand kilometres, and a round trip takes roughly a hundred and twenty milliseconds. Light is the fastest thing in the universe — so even across the planet, it feels immediate.

[CUE] Reveal the Mumbai → London stat card as you say the numbers.

08 Why it matters

And once you get this, a lot of things click. A server far away means more distance, which means more delay — that's **ping**, or lag, in your games. "The cloud" is just servers in big buildings, joined by these cables. And if one undersea cable gets cut, an entire region's internet can slow down.

[CUE] Relate to gaming lag — your audience feels this one.

09 Common myth

So the myth: people think the internet is wireless — floating through the air, beamed by satellites. The reality: about ninety-nine percent is **physical cable**, including undersea fibre between continents. The wireless part — WiFi, mobile data — is only the **last hop** to you.

[CUE] Reveal the green "Reality" card after a beat on the myth.

10 Recap

Quick recap. One: the internet is a giant web of **physical cables** — wireless is just the last few metres to you. Two: your data races as **light** to a server's address and back — that's the millisecond magic.

[CUE] Crisp. Two clean lines.

11 CTA

So the internet is really cables plus light — wrapping the whole planet. If that blew your mind a little, subscribe to Katixo KhojLab. Next question: what **really** happens in the moment you click "Send"? See you there.

[CUE] Warm, friendly. Smile on the subscribe line.
